

Technology Stack

Date	27 JUNE 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, JavaScript, Streamlit	Streamlit provides a fast and interactive interface, while HTML, CSS, and JavaScript improve the user interface and responsiveness.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, FastAPI	FastAPI provides high-performance REST APIs and integrates easily with AI/ML models for processing user requests.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	Firebase Firestore	Firestore securely stores user profiles, learning history, quiz results, and progress with real-time synchronization.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Google Cloud Platform (GCP), Streamlit Cloud	GCP provides scalable cloud infrastructure for AI services, while Streamlit Cloud enables easy deployment of the application.
5	Version Control & CI/CD	Git, GitHub	Git and GitHub are used for version control, collaboration, project backup, and source code management.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Google Gemini API, Firebase Authentication, Google Translate API</i>	Gemini API powers AI tutoring and content generation, Firebase Authentication manages secure login, and Google Translate API supports multilingual learning